

Relative complex permittivity and its dependence on frequency

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Abstract

Purpose – This paper aims to give a brief overview of dielectric properties, relative complex permittivity and its dependence on frequency. The significance of different approaches to complex permittivity is also discussed.

Design/methodology/approach – The different mechanisms of polarization are then presented. Dielectric measurements are given, and an RC parallel-equivalent circuit is used to simulate a parallel plate capacitor, and the way in which the impedance of the circuits is affected by frequency is illustrated in their respective diagrams. The way in which dielectric properties change with time is also discussed.

Findings – The goal of this paper is to give an overview of the characteristics of the dielectrics and how frequency affects the relative complex permittivity and to present different approaches to and equations for the relative complex permittivity, such as that of Debye, Cole–Cole, Cole–Davidson and Havriliak–Negami. In addition, three mechanisms of polarization, namely, electronic, atomic and bipolar, are presented. The most common dielectric characterization device, a capacitor with parallel plates between which the dielectric material under study is located, is also discussed. Ohmic and dielectric losses of a non-ideal capacitor are accounted for. Furthermore, this paper studies the equivalent circuits of a non-ideal parallel plate capacitor, those being a resistor and an ideal capacitor connected either in series or in parallel.

Originality/value – Finally, dielectric responses to both time and to stepwise excitation are given.

Keywords Frequency, Cole–Cole, RC parallel circuit, Relative complex permittivity

Paper type General review

Introduction

When an electric field (EF) can be maintained with zero or almost-zero power losses in a material, it is characterized as a dielectric or an electrical insulator. In fact, the dielectric is not an ideal insulator because a number of electrons can pass through the material. Because of the processes of changing the polarity in the material, a part of the electrical energy is lost as heat. The electric force between the charges in a material is given by Coulomb's law (Grimnes and Martinsen, 2015):

$$F = \frac{1}{4\pi\epsilon} \frac{Q_1 Q_2}{r^2} \quad (1)$$

where $\epsilon_0 = 8.854 \cdot 10^{-12}$ F/m is the electric permittivity of free space; ϵ is the absolute permittivity of the medium, which is given as $\epsilon = \epsilon_r \times \epsilon_0$; and ϵ_r is the relative permittivity of the medium. The dimensions of ϵ and ϵ_0 are F/m in SI, but ϵ_r is a dimensionless number greater or equal to 1 in the case of the vacuum.

The capacitance of a capacitor consisting of two parallel plates is given by equation (2):

$$C = \frac{\epsilon_0 A}{d} \quad (2)$$

where A is the area of each plate and d is the distance between them.

The presence of a dielectric material between the plates reduces the electrical field between the plates owing to the existence of a polarized field in the opposite direction within the material, in which case, the above equation becomes:

$$C = \frac{\epsilon A}{d} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d} \text{ or } C = \epsilon_r C_0 \quad (3)$$

For an ideal dielectric material, ϵ is a real number, but in practice for any dielectric material, there is also an imaginary component that is associated with the dielectric loss because of the poor conductivity of the material. Therefore, ϵ is called the absolute complex permittivity, which is also denoted by ϵ^* and describes the total interaction of the dielectric materials with the variable electrical field. In this case, the following equation is used:

$$\epsilon = \epsilon' - j\epsilon'' \quad (4)$$

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Relative complex permittivity (ϵ_r)

In practice, the relative complex permittivity is used instead of the absolute complex permittivity, because it is dimensionless and takes simple numerical values. It is equal to the absolute complex permittivity divided by the electrical permittivity of free space (Von Hippel, 1995); therefore:

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = \left(\frac{\epsilon'}{\epsilon_0} \right) - j \left(\frac{\epsilon''}{\epsilon_0} \right) = \epsilon_r' - j\epsilon_r'' \quad (5)$$

ϵ' is the real part of the relative complex permittivity, which is related to the stored energy within the medium.

ϵ_r'' is the imaginary part of the relative complex permittivity, which is related to the dissipation (or loss) of energy within the medium.

The ϵ_r' is an indication of how much energy can be stored in the material by the applied EF and is called relative permittivity. The ϵ_r'' is related to the loss of energy in the medium and is an **indication of how polar the loose material is to the external imposed EF**. The loose material in a frequency occurs when the polarization mechanism of the material is able to follow the ER changes applied to the material. The ϵ_r'' is always a positive quantity much smaller than the ϵ_r' .

When the relative complex dielectric permittivity ϵ_r is designed in a vector format, the real and imaginary component of the equivalent circuit must have a phase difference of 90° . The vector sum forms the angle δ with the real axis ϵ_r' (Figure 1). The tangent of this angle $\tan\delta$ is an important factor in characterizing dielectrics and is called the tangent loss.

The currents' charges and losses are proportional to the energy stored in the capacitance and the energy released in the form of heat, respectively. The expected levels of relaxation of any material are derived from the ratio of the emitted and stored energy in a period of time. For each cycle, the following applies:

$$\tan \delta = \frac{\epsilon_r''}{\epsilon_r'} \quad (6)$$

For good insulating materials, the vector ϵ_r approaches that of ϵ_r' , while $\tan\delta \rightarrow 0$.

Dependence of dielectric permittivity on frequency

The dependence of the frequency of the dielectric permittivity is associated with the mechanisms of polarization, which take place in each frequency range (Figure 2). At low frequencies, all mechanisms are present. Increasing the frequency and reaching

Figure 1 Vector diagram of the relative complex dielectric permittivity

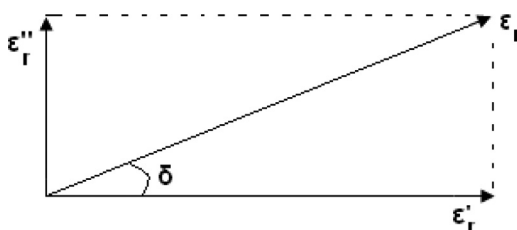
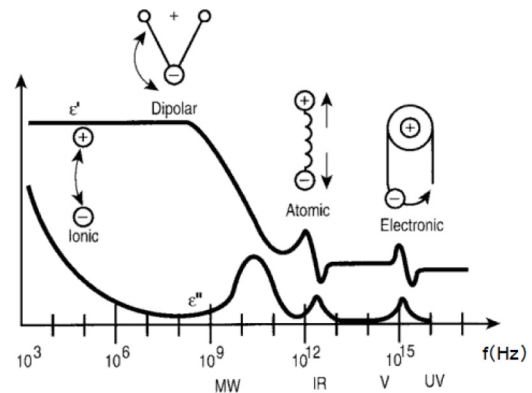


Figure 2 ϵ' and ϵ'' change with frequency



Source: Levy (1968)

the microwave region, the permanent dipoles, owing to inertia, can monitor changes in the field and align with it. The polarization orientation stops and the dielectric permittivity declines. Also, some energy no longer goes to the circuit but is absorbed by the material, indicating the existence of losses. The curve ϵ'' in this region takes the form of a resonance curve. At higher frequencies in the infrared, the mechanism of ionic polarization stops; at higher frequencies in the UV, the mechanism of electronic polarization stops.

The real part of dielectric permittivity ϵ' varies between a maximum value ϵ_s , corresponding to static fields or very low frequencies and a minimum value ϵ_∞ , corresponding to very high frequencies. The imaginary part ϵ'' is related to losses and has a maximum at the resonance frequency.

The ϵ' and ϵ'' are given by Debye's equations (Debye, 1929):

$$\epsilon' = \epsilon_\infty + \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2}, \quad \epsilon'' = \omega \tau \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2} \quad (7)$$

where τ is the relaxation time of the material.

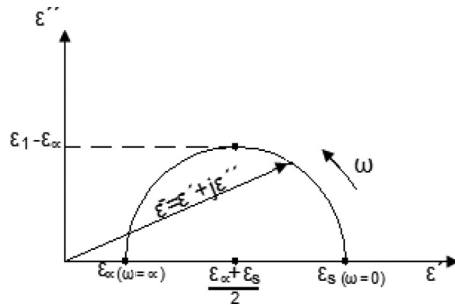
From the above equations, we get:

$$\begin{aligned} \epsilon - \epsilon_\infty &= \epsilon' - \epsilon_\infty - j\epsilon'' = \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2} - j \omega \tau \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2} \\ \Rightarrow \epsilon'^2 + \epsilon''^2 &= \epsilon'(\epsilon_s' - \epsilon_\infty) - \epsilon_s \epsilon_\infty \end{aligned} \quad (8)$$

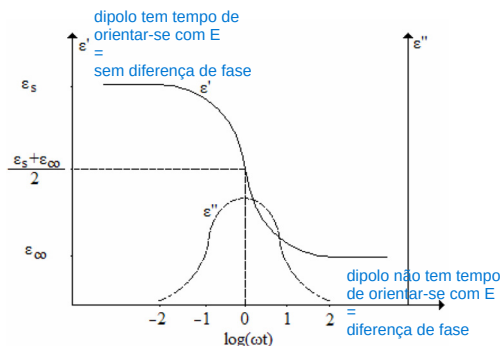
Equation (8) in a diagram representing a circle has its center on the axis ϵ' at the point $(\epsilon_s + \epsilon_\infty)/2$. Figure 3 depicts the Cole-Cole mechanism graph.

A dielectric material subjected to a field can be visualized at any frequency from an equivalent circuit of a capacitance and a resistance in either series or in parallel. For materials that exhibit dielectric losses, the parallel equivalent is usually considered more suitable.

As shown in Figure 4, in a frequency range around the value $\omega = 1/\tau$, we observe a peak in ϵ'' ; this is because in this region energy losses are maximized owing to the frequencies ω and $1/\tau$ being comparable. The maximum peak is located at the frequency:

Figure 3 Cole–Cole graph of the complex permittivity ε 

Source: Levy (1968)

Figure 4 Real and imaginary part of the dielectric permittivity for a Debye mechanism

Source: Levy (1968)

$$\omega_{\max} = 1/\tau \quad (9)$$

We also see a step in ε because at much smaller frequencies than $\omega_{\max}$, the dipoles have time to follow the changes in the field, whereas at much higher frequencies, they do not; thus, they are not involved in polarization at low frequencies $\varepsilon' = \varepsilon_s$ and at high frequencies $\varepsilon' = \varepsilon_\infty$.

Debye's dispersion equations describe a relaxation process, which is characterized by a single relaxation time, and each relaxation process is described separately. The experimental results are consistent with the theoretical results in the case of polar liquids. However, when studying systems in condensed matter, owing to interactions between atoms and molecules, several relaxation times appear and the experimental data deviate from Debye's theory. For a description of the experimental data, different theories are used such as the Cole–Cole, Cole–Davidson and Havriliak–Negami because they take into account different types of distribution of relaxation times (Runt and Fitzgerald, 1997). The expression of the complex permittivity for each approach [Debye (1929), Cole and Cole (1941), Davidson and Cole (1950), Navriliak and Havriliak (1997)] is given by the following relations:

Debye:

$$\varepsilon = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)(1 - j\omega\tau)}{1 + \omega^2 \tau^2} \quad (10)$$

Cole–Cole:

$$\varepsilon = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)}{1 + (j\omega\tau)^{1-\alpha}}, \quad 0 \leq \alpha \leq 1 \quad (11)$$

where α is a parameter and the distribution of relaxation times is symmetrical around τ . If $\alpha = 0$, then it gives the Debye equation.

Cole–Davidson:

$$\varepsilon = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)}{(1 + j\omega\tau)^\beta}, \quad 0 \leq \beta \leq 1 \quad (12)$$

The distribution of relaxation times is not symmetrical and β is a parameter that determines the shape. If $\beta = 1$, it gives the Debye equation.

Havriliak–Negami (Hill *et al.*, 1969; Boettcher and Bordewijk, 1992; Havriliak and Negami, 1966):

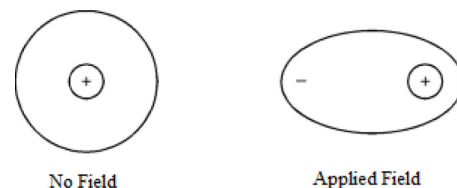
$$\varepsilon = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)}{[1 + (j\omega\tau)^{1-\alpha}]^\beta}, \quad 0 \leq \alpha \leq 1 \text{ and } 0 \leq (1-\alpha)\beta \leq 1 \quad (13)$$

if $\beta = 1$, it gives the Cole–Cole equation; if $\alpha = 0$, it gives the Cole–Davidson equation; and if $\beta = 1$ and $\alpha = 0$, it gives the Debye equation (Hedvig, 1977).

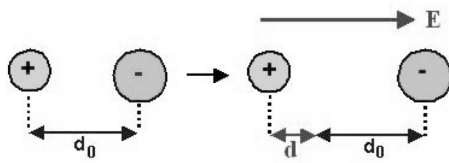
Dielectric polarization

There are different mechanisms of polarization, with the main ones being electronic (Figure 5), ionic (Figure 6), bipolar and atomic (Blythe and Bloor, 2005):

- The electronic polarization effect on each atom or molecule as the center of mass of the electron cloud that surrounds the atom will shift because of the EF. This effect is extremely fast, as the electrons are very light and can be observed by optical frequencies.
- Atomic or ionic polarization refers to materials whose molecules form ions. When an EF is applied, cations and anions become displaced in opposite directions. Apart from electric polarization, elastic displacements of the electrical charges (nuclei and electrons) will also occur. This means that these types of molecules are polar

Figure 5 Electronic polarization

Source: www.tf.uni-kiel.de/matwis/amat/admat_en/index.html (accessed 10 May 2016)

Figure 6 Ionic polarization

Source: www.tf.uni-kiel.de/matwis/amat/admat_en/index.html (accessed 10 May 2016)

compounds that can be polarized up to infrared frequencies.

- Bipolar polarization refers to materials that contain molecules with permanent dipole moments. Under the influence of EF, the dipoles are oriented in the direction of the EF.

In conclusion, the dielectric polarization is the result of the relative displacement of positive and negative charges in a material. During this process, the EF is not able to force the charge to escape from the material which would cause electrical conductivity. Each of the dielectric polarization mechanisms is associated with a characteristic resonance frequency (sharp increase of dielectric permittivity at a given frequency) and relaxation frequency (gradual decrease in dielectric permittivity with increasing frequency). As the frequency increases, the slower dielectric polarization mechanisms are eliminated, leaving only the fastest to contribute to the phenomena of energy storage (ϵ_r'). Similarly, the rate of loss (ϵ_r'') is incremented at each critical frequency. The resonance is usually associated with electronic polarization.

Parallel plates with a dielectric

The most common dielectric characterization device is a capacitor with parallel plates between which the dielectric material under study is located.

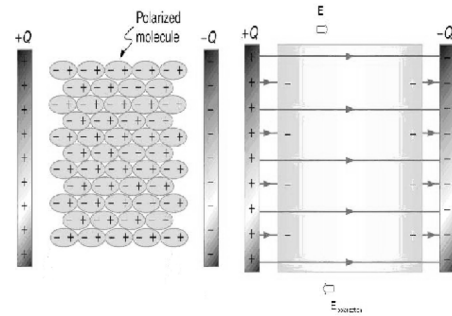
The capacity of a pair of parallel charged plates is increased by introducing a dielectric material. The capacitance is inversely proportional to the EF, while the presence of the dielectric reduces the field between the plates owing to the existence of a field with opposite direction within the material because of the polarization of the material (Figure 7).

The capacity of a pair of parallel charged plates increases when a dielectric material is introduced. It is known that the dielectric is characterized by the dielectric permittivity ϵ_r and the capacity is multiplied by this factor. Overview of the capacity is:

$$C = \frac{\epsilon_r \cdot \epsilon_0 \cdot A}{d} \quad (14)$$

The EF when there is air or a dielectric material between the plates is:

$$E = \frac{\sigma}{\epsilon} = \frac{V}{d} \quad (15)$$

Figure 7 Parallel plates charged

Notes: Polarized molecules (left) and EF (right)

Source: <https://cnx.org/contents/VtjKSKsR@4/Capacitors-and-Dielectrics> (accessed 10 May 2016)

and:

$$E_{\text{effective}} = E - E_{\text{polarization}} = \frac{\sigma}{\epsilon_r \cdot \epsilon_0} \quad \text{respectively,} \quad (16)$$

where V is the applied voltage and d the distance between the plates and σ is the surface charge density in each plate. The relative dielectric permittivity ϵ_r shows the amount of reduction of the active field in relation to the initial field. This reduction of the field results in the ability for more energy to be stored in the capacitor having the same voltage between the plates.

The electrical equivalent circuit which is used to simulate a parallel plate capacitor may consist of a resistor and an ideal capacitor connected parallel (Figure 8).

The vector diagram of voltage and current for the equivalent circuit is shown in Figures 9 and 10, respectively.

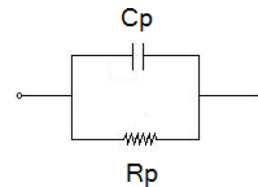
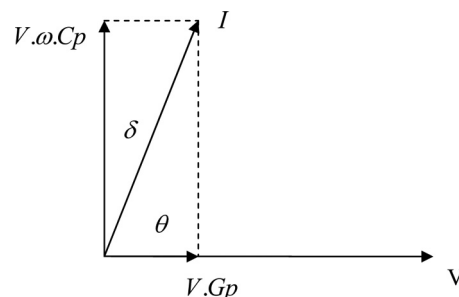
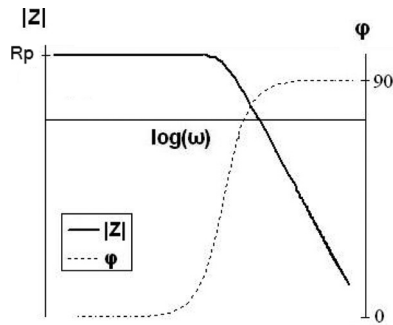
Figure 8 The RC parallel circuit**Figure 9** Vector diagram of voltage and current of the parallel circuit

Figure 10 Frequency responses (Bode diagrams) of the parallel electrical circuit

The loss tangent from the parallel circuit is:

$$\tan \delta = \frac{1}{\omega C_p R_p} = \frac{\varepsilon_r''}{\varepsilon_r'} \quad (17)$$

The complex impedance of the parallel circuit is:

$$Z_p = \frac{1}{\frac{1}{R_p} + j\omega C_p} \quad (18)$$

The real and the imaginary part of dielectric permittivity is:

$$\varepsilon_r' = \frac{C_p}{C_0}, \quad \varepsilon_r'' = \varepsilon_r' \tan \delta \quad (19)$$

Finally, the complex permittivity is:

$$\varepsilon_r = \varepsilon_r' - j\varepsilon_r'' \quad (20)$$

In the above analysis, we assumed that the electrical field which is applied to the dielectric is sinusoidal with a stable frequency. The dielectric spectroscopy studies the changes in dielectric properties of the material with time and frequency. Inside an isotropic and homogeneous dielectric, the polarization density vector P and the intensity E of the EF have the same direction and are linked in the equation (Frubing, 2001; Froehlich, 1958):

$$P = \chi \varepsilon_0 E \quad (21)$$

where χ is the electrical susceptibility of the material. The χ indicates the degree of all kinds of polarizations of a dielectric and it is dimensionless (for the vacuum, it is equal to zero). The $\varepsilon_0 = 8.85419 \times 10^{-12}$ As/Vm is the dielectric constant of the vacuum. The electric displacement D is determined by the overall positive or negative electric charge per unit area induced in the corresponding electrode. The electrical charge of the two electrodes is the origin of all the electrical field lines. In the area between the electrodes without a dielectric, the electrical displacement is parallel to the EF E and is connected to it by the equation $D = \varepsilon_0 E$. When there is an isotropic dielectric between the electrodes, the electrical displacement increases by the polarization density P and the equation becomes $D = \varepsilon_0 E + P$

(Barsoukov and Macdonald, 2005; Boettcher, 1993). This happens because in each electrode, a part of the charge creates the EF E , while the remaining charge compensates the polarization charges of the dielectric. Considering the above equations, we get:

$$D = \varepsilon_0 E + \chi \varepsilon_0 E = (1 + \chi) \varepsilon_0 E \quad (22)$$

Thus, in a time constant electrical field, the electrical displacement D is proportional to E , and for isotropic dielectric materials, the vectors P and D are parallel to E . The proportionality factor $(1 + \chi) \varepsilon_0$ is the dielectric permittivity ε of the material and the $1 + \chi$ is the relative dielectric permittivity ε_r . In this case, the use of complex equations is not necessary.

When the electrical field is the time-varying $E(t)$, then the D in the vacuum immediately follows changes in the field $D(t) = \varepsilon_0 E(t)$. The charge density at the electrodes is determined by the displacement current, which is derived from the voltage source and equals dQ/dt , where Q is the total charge on each electrode. When the vacuum is replaced by an isotropic dielectric material, the electric displacement D is:

$$D(t) = \varepsilon_0(t) + P(t) \quad (23)$$

The time dependence of $P(t)$ is not the same as that of $E(t)$ because the polarization of the dielectric is not directly related to the applied field, but it has a time delay which is different for the various types of polarizations. To find the relationship between the field and the polarization, we will try to determine a time-varying function of the electrical susceptibility $\chi = \chi(t)$ and, hence, the relative dielectric permittivity $\varepsilon_r = 1 + \chi(t)$ applies to the equations:

$$P(t) = \chi(t) \varepsilon_0 E(t) \quad (24)$$

and:

$$D(t) = (1 + \chi(t)) \varepsilon_0 E(t) \quad (25)$$

Summary

The goal of this paper was to give an overview of the characteristics of dielectrics and how frequency affects relative complex permittivity and to present different approaches to and equations for relative complex permittivity, such as that of Debye, Cole–Cole, Cole–Davidson and Havriliak–Negami. The above study is within the framework of non-equilibrium thermodynamics (de Groot and Mazur, 1984). In addition, three mechanisms of polarization, namely, electronic, atomic and bipolar, are presented. The most common dielectric characterization device, a capacitor with parallel plates between which the dielectric material under study is located, is also discussed. Ohmic and dielectric losses of a non-ideal capacitor are accounted for. Furthermore, we studied the equivalent circuits of a non-ideal parallel plate capacitor, those being a resistor and an ideal capacitor connected either in series or parallel. Finally, dielectric responses to both time and to stepwise excitation are given.

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